A PROJECT TITLE

ON

Precision-Farming-Using-Drone-Imagery-and-CNN-Based-Crop-Analysis

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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1. Abstract:

Agriculture is one of the most important sectors for ensuring food security and sustainable development. Traditional methods of crop monitoring require significant time, labour, and manual effort, making it difficult to identify crop diseases at an early stage. This project presents a precision farming system that uses drone imagery and Convolutional Neural Networks (CNNs) to analyse crop health accurately and efficiently.

In this system, drones equipped with high-resolution cameras capture aerial images of agricultural fields. The collected images are processed using image preprocessing techniques such as resizing, noise removal, and normalization. A CNN model is then trained to classify healthy and diseased crops based on the extracted features from the images. The model can identify crop stress, disease symptoms, and variations in plant growth with high accuracy.

The proposed system enables farmers to monitor large agricultural areas quickly and detect diseases at an early stage, reducing crop losses and improving productivity. It also provides valuable insights for irrigation management, fertilizer application, and pest control, thereby reducing unnecessary resource usage and promoting sustainable farming practices.

The system is designed using Python, TensorFlow, OpenCV, and Jupyter Notebook. The trained CNN model predicts crop conditions in real time, allowing farmers to make informed decisions based on accurate data. The integration of drone technology and artificial intelligence significantly reduces manual inspection costs while increasing agricultural efficiency.

Overall, this project demonstrates how modern technologies such as drones and deep learning can transform traditional farming into smart agriculture. It improves crop health monitoring, enhances yield prediction, minimizes production costs, and supports precision farming practices for future agricultural development.

2. INTRODUCTION:

Precision farming is an advanced agricultural approach that uses modern technologies to improve crop productivity and resource management. Traditional crop monitoring methods rely on manual field inspections, which are time-consuming and may not detect diseases at an early stage.

This project uses drones to capture aerial images of agricultural fields. These images are analysed using Convolutional Neural Networks (CNN), a deep learning technique that automatically detects crop diseases and evaluates plant health. The system provides accurate information about crop conditions, enabling farmers to take timely actions such as irrigation, pesticide spraying, and fertilizer application.

The proposed system helps reduce labour costs, improve crop yield, save water, minimize chemical usage, and support sustainable agriculture through intelligent decision-making.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11 or Ubuntu Linux
- Programming Language: Python 3.10+
- IDE: Jupyter Notebook / Visual Studio Code
- Libraries:
  - TensorFlow / Keras
  - OpenCV
  - NumPy
  - Pandas
  - Matplotlib
  - Scikit-learn
- Dataset: Drone Crop Image Dataset (Healthy and Diseased Crop Images)
- Hardware:
  - Intel Core i5/i7 Processor
  - 8 GB RAM or above
  - 20 GB Free Storage
  - GPU (Optional, for faster CNN training)

Signature of the Guide: **Course Instructor:**